# Metrics Match Consequences

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

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## Metrics Match Consequences

> Prepare Metrics Match Consequences: check the target, prerequisites, resources and adaptable teaching schedule.

### Teach the learning cycle

Primary target: Apply a declared **score threshold** including equality

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

### Retrieval answer guidance

### Train, Validation, Test

Without opening the old lesson, explain one key idea from Train, Validation, Test. Give an example and a mistake that the idea helps you avoid.

Look for: Select a model using a declared validation rule

Reference example, not the only acceptable wording: Teacher explanation: Choose A: validation MAE 3 ties C, and A wins the simplicity tie. B has validation MAE 5 despite **training** MAE 0.

### Train a Regression Model

Without opening the old lesson, explain one key idea from Train a Regression Model. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a line from a declared **training** candidate set

Reference example, not the only acceptable wording: Teacher explanation: A 5 B 3 C 0; choose m 2 b 1.

### Regression and Residuals

Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish numeric **prediction** from classification

Reference example, not the only acceptable wording: Teacher explanation: Residuals -1, 1, 0 cm; the first record is overpredicted.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

### What the learner must demonstrate

> A reviewer can inspect at most 10 alerts daily. Model A flags 20 cases with higher recall; B flags 8 with lower recall. What **evidence** is needed before choosing a **threshold**?

### Concept to explain

Metric choice depends on intended use, the positive class, who is affected, and the **consequence**s of each error.

| Term | Meaning |
|---|---|
| consequence | real effect of a prediction error |
| threshold | score boundary for a class decision |
| cost matrix | relative impact assigned to error types |
| metric suite | multiple measures covering distinct concerns |

### Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

## Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

> Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

### Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on **threshold**s and **consequence**s. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

### Readiness answer

All reference positives, TP+FN.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Optional Python 3 editor; no external packages or AI accounts required

### End goal for the pictorial case

By the end, I can explain metrics match **consequence**s using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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## Explain the mechanism

> Model Metrics Match Consequences using the supplied case, labelled mechanism and

> reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and evaluation proof separately.

### Question to explain

QUESTION Select metrics for screening, recommendation, and forecasting tasks.

MODEL IDEA Metric choice depends on intended use, the positive class, who is affected, and the **consequence**s of each error.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

### How the mechanism works

UNDER THE HOOD Screening task treats missed positives as costly, so emphasize recall and human follow-up. Recommendation task may track precision among shown items plus user benefit.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

### Step 1

STEP 1 Screening task treats missed positives as costly, so emphasize recall and human follow-up.

### Step 2

STEP 2 Recommendation task may track precision among shown items plus user benefit.

### Step 3

STEP 3 Height forecast uses MAE in centimeters and residual slices.

### Step 4

STEP 4 No metric replaces a written **consequence** analysis and subgroup review.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a **score**.

Recalculate, retrace, or restate the **evidence** independently:

**Concrete teaching case**

### The teaching case

Spotting alerts can miss disease signs or create unnecessary healthy-leaf reviews.

### The teaching case

Connect metrics to **consequence**s.

**Exact explanation to model**

### Why the result follows

Recall measures finding positives; precision measures alert correctness. Set tradeoffs using **consequence**s, review capacity and validation **evidence**.

**Explain the added worked example**

## Explain the added worked example · 1

A classifier **score** is not automatically a probability. Declare blocked positive and predict blocked when **score** is at least the **threshold**. Changing this boundary changes decisions without necessarily re**training** the model.

Select it using development **evidence** and a task-relevant rule, not final-test answers.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **threshold**s and **consequence**s.

Six records have **score**/reference pairs A .9/blocked, B .7/clear, C .6/blocked, D .4/blocked, E .3/clear, F .1/clear. At **threshold** .5, TP=2, FP=1, FN=1, TN=2. At .3, TP=3, FP=2, FN=0, TN=1.

Equality is positive, so E is flagged at .3.

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

### Key terms

| Term | Meaning |
|---|---|
| Score | A model output used to rank or decide |
| Threshold | A boundary that converts a score to a label |
| False alarm | A positive prediction for a negative reference |
| Cost rule | A declared way to value different mistakes |

### Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

### Thresholds and consequences at a glance

| PART | EVIDENCE OR ACTION | INTERPRETATION |
|---|---|---|
| Threshold .5 | One false alarm one miss | Four of six correct |
| Threshold .3 | Two false alarms no misses | Four of six correct |
| Decision rule | Include equality | Record before evaluation |

Lowering the threshold here finds D but adds E as a false alarm. The two accuracies are equal while consequences differ. Precision is 2/3 at .5 and 3/5 at .3; recall rises from 2/3 to 1. This behavior follows these fixed scores and records, not a promise of future performance.

**Give TP, FP, FN, TN at both thresholds. Explain the steps or evidence behind your response.**

| Thinking move | Teacher prompt |
|---|---|
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |

| Thinking move | Teacher prompt |
| --- | --- |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

### A pictorial worked case: Metrics Match Consequences

Translate a **threshold** choice into missed-leaf and false-alarm costs.

### The facts we will use

#### Supplied facts

**Score**s/reference: 0.9 spotted, 0.6 healthy, 0.4 spotted, 0.1 healthy. Predict spotted at **score**>=**threshold**. Miss cost5; false-alarm cost1.

#### Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/25

# Metrics Match Consequences

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Threshold0.7

TP 1, FN 1, FP 0, TN 2: cost5.



## 2. Threshold0.4

TP 2, FN 0, FP 1, TN 1: cost1.



## 3. Choose for stated costs

Threshold0.4 has smaller validation cost on these four cases.

**CHANGE ONE THING**

Miss and false-alarm costs swap to1 and 5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Threshold0.7 | TP 1, FN 1, FP 0, TN 2: cost5. |
| 2. Threshold0.4 | TP 2, FN 0, FP 1, TN 1: cost1. |
| 3. Choose for stated costs | Threshold0.4 has smaller validation cost on these four cases. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows


### Why the result follows


Read the picture in order. **Threshold**0.4 has smaller validation cost on these four cases. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

### What this example does not establish

### Keep the stated limit

Toy costs illustrate decision design; real use needs agreed **consequence**s, review capacity and fresh evaluation.

### Change one thing in the pictured case

#### Change one thing

Change: Miss and false-alarm costs swap to1 and 5.

#### Predicted result

**Prediction**: **Threshold**0.7 costs1 while0.4 costs5; preferred **threshold** switches.

#### Why the result follows

Reason: Metric choice encodes the declared **consequence**s, not a universal best **threshold**.

### Ask while showing the picture

#### Ask while showing the picture · 2

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

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## Demonstrate and guide practice

> Demonstrate Metrics Match Consequences, reduce support deliberately and check individual reasoning before discussing answers.

## Board sequence

### 1. Put the facts on the board

Write the given facts: Spotting alerts can miss disease signs or create unnecessary healthy-leaf reviews.

### 2. State the decision to explain

Write the question to explain: Connect metrics to **consequence**s.

### 3. Reveal the worked reasoning

Reveal after **prediction**s: Recall measures finding positives; precision measures alert correctness. Set tradeoffs using **consequence**s, review capacity and validation **evidence**.

### Hold back the independent task

Keep for an independent attempt: Missing a diseased plant is considered more costly than a **false alarm**, but staff can review only ten alerts daily. Which evaluation trade-off should be discussed?

## Classroom dialogue

### Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

### Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

### Model these guided questions

Guided response: At .5: 2, 1, 1, 2. At .3: 3, 2, 0, 1. Exit response: Selection uses the test answers, turning the test into development data and weakening its independent evaluation role.

### Case-specific teaching sequence

Stage · Teacher action

> **Stage:** Session 1 start to 7 minutes
>
> **Teacher action:** Use the prerequisite question: Which denominator defines recall? Check the explanation before proceeding.

> **Stage:** 7 to 18 minutes
>
> **Teacher action:** Read the opening and model the worked evidence. Ask students to identify score in context.

> **Stage:** 18 to 32 minutes
>
> **Teacher action:** Complete W 1 and W 2 with support: apply the boundary; count the consequences. Require the evidence, not only the final response.

> **Stage:** 32 to 45 minutes

**Teacher action:** Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Trace the Python check after the paper calculation.

**Stage:** Session 2 start to 8 minutes

**Teacher action:** Retrieve the distinction between score and threshold. Invite a counterexample to an overconfident claim.

**Stage:** 8 to 20 minutes

**Teacher action:** Discuss the application and complete W 3: Calculate costs with FP cost 1 and FN cost 4.

**Stage:** 20 to 35 minutes

**Teacher action:** Collect W4 independently before giving hints. Assess these outcomes: Apply a declared score threshold including equality; Compare false alarms and misses at two thresholds. Record any support used.

**Stage:** 35 to 45 minutes

**Teacher action:** Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

### Reduce help deliberately

#### Completion practice

Completion task: Give TP, FP, FN, TN at both **threshold**s.

#### Reasoning to listen for

Teacher check: Teacher explanation: At .5: 2, 1, 1, 2. At .3: 3, 2, 0, 1.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

### Individual reasoning check

Use the worked case for **Threshold**s and **consequence**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> ### Expected repair
>
> Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: At .5: A, B, C. At .3: A, B, C, D, E, including equality.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

### Pictorial practice question

Which labels determine the false-alarm count at **threshold** 0.4? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

### Reasoned teacher answer

The healthy 0.6 case is wrongly flagged; healthy 0.1 stays clear. There is exactly one **false alarm**. Changed-case check: **Threshold**0.7 costs1 while0.4 costs5; preferred **threshold** switches.

Metric choice encodes the declared **consequence**s, not a universal best **threshold**.

Changed-case reasoning: **Threshold**0.7 costs1 while0.4 costs5; preferred **threshold** switches. Metric choice encodes the declared **consequence**s, not a universal best **threshold**.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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TEACHER GUIDE · 4 OF 9

## Run the weekly evidence lab

> Run the Metrics Match Consequences investigation and use the first unsupported step to choose a teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected **evidence** before running.

2. Metric Council: map stakeholders and error costs, select a **metric suite**, and defend what each metric cannot show.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

### Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

| Record | What to capture |
| --- | --- |
| Input or evidence | Exact data, instruction or claim |
| Expected result | Answer or safe fallback declared before the run |
| Observed result | Actual result, including errors |
| Explanation | Rule, calculation or evidence explaining the outcome |
| Next test | One justified change and a new prediction |

### Diagnose and reteach

MISCONCEPTION TO REPAIR The same highest-accuracy metric should choose every model.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

### Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: At .5: A, B, C. At .3: A, B, C, D, E, including equality.

A stronger claim makes a stronger project: Selection uses the test answers, turning the test into development data and weakening its independent evaluation role.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: At .5 TP 1 FP 1 FN 1 TN 1 cost 5. At .2 TP 2 FP 1 FN 0 TN 1 cost 1. Equality is positive at both boundaries.

### Match the support to the first error

| What the response shows | Next teaching action |
|---|---|
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

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## Assess and explain the answers

> Use the Metrics Match Consequences questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

### Worked case answer

#### Teacher key · Worked case answer

Recall measures finding positives; precision measures alert correctness. Set tradeoffs using **consequence**s, review capacity and validation **evidence**.

### Independent question

#### Independent question · withhold the key

Missing a diseased plant is considered more costly than a **false alarm**, but staff can review only ten alerts daily. Which evaluation trade-off should be discussed?

### Expected independent answer

#### Teacher key · Expected independent answer

Prioritize disease recall while checking precision and alert volume against review capacity. Choose **threshold**s using validation **evidence** and the stated **consequence**s.

### What counts as evidence

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Result | Incorrect or absent | Partly correct | Correct with necessary conditions |
| Reasoning | No supporting trace | Some steps | Clear mechanism, calculation or source support |
| Limit | Overclaims certainty | Vague limitation | Relevant uncertainty or missing fact |
| Transfer | Copies without adapting | Adapts with prompts | Solves a changed case independently |

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification **threshold**.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

### Feedback

Ask the learner to find the **evidence** that challenges this claim: “One metric can be maximized without considering workload or other error types.” Keep the first attempt and an uncoached retry separate.

### Original transfer answer

#### Teacher key · Original transfer answer

Compare validation precision/recall across **threshold**s, missed-positive **consequence**s and review workload under realistic volume. More recall can overload review; fewer alerts can miss important cases.

Choose a documented tradeoff before final testing rather than treating one metric as universally best.

### Explanations for the additional workbook examples

### W1 Apply the boundary

#### Workbook question · W1

Which IDs are positive at .5 and .3?

#### Teacher reasoning · W1

Teacher explanation: At .5: A, B, C. At .3: A, B, C, D, E, including equality.

### W2 Count the consequences

#### Workbook question · W2

Give TP, FP, FN, TN at both **threshold**s.

#### Teacher reasoning · W2

Teacher explanation: At .5: 2, 1, 1, 2. At .3: 3, 2, 0, 1.

### W3 Use the declared cost

#### Workbook question · W3

Calculate costs with FP cost 1 and FN cost 4.

#### Teacher reasoning · W3

Teacher explanation: Costs are 1+4=5 and 2+0=2; .3 is preferred among these candidates.

### W4 Apply independently

#### Workbook question · W4

**Score**s/references are .8/blocked,.5/clear,.2/blocked,.1/clear. Compare **threshold**s .5 and .2 using the same cost.

#### Teacher reasoning · W4

Teacher explanation: At .5 TP 1 FP 1 FN 1 TN 1 cost 5. At .2 TP 2 FP 1 FN 0 TN 1 cost 1. Equality is positive at both boundaries.

### W5 Protect the final test

#### Workbook question · W5

Why not select the **threshold** with the lowest final-test cost?

#### Teacher reasoning · W5

Teacher explanation: Selection uses the test answers, turning the test into development data and weakening its independent evaluation role.

### Assess independent transfer

#### Independent question · withhold the key

Selected case: **Score**s/references are .8/blocked,.5/clear,.2/blocked,.1/clear. Compare **threshold**s .5 and .2 using the same cost.

### Selected case answer

#### Teacher key · Selected case answer

Teacher explanation: At .5 TP 1 FP 1 FN 1 TN 1 cost 5. At .2 TP 2 FP 1 FN 0 TN 1 cost 1. Equality is positive at both boundaries.

| Dimension | Evidence required |
|---|---|
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

### Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

### Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

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## Connect and extend learning

> Connect Metrics Match Consequences to Responsible Plant Classifier, adapt support and plan a later changed-case check.

### Responsible Plant Classifier

Approve the Plant Classifier **metric suite**: accuracy for context, spotted precision/recall, confusion matrix, and subgroup slices.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

### Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

### Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

### Bridge to the next week

Next: Week 26: Reproducible Experiments. Retrieve today's **evidence** and identify what carries forward and what is new.

### Support and fresh reassessment

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **score** and **threshold** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: At .5 TP 1 FP 1 FN 1 TN 1 cost 5. At .2 TP 2 FP 1 FN 0 TN 1 cost 1. Equality is positive at both boundaries.

Fresh retry .9/clear,.6/blocked,.4/clear,.2/blocked at **threshold** .4: TP 1 FP 2 FN 1 TN 0, cost 6 under FP 1 FN 4. Require equality and all counts. Mark each criterion as independent, supported or not yet demonstrated.

Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Calibration as a group property. Carry forward the saved **evidence** and the distinction between development choices and justified claims.

### Plan the delayed return

Suggested return points in this level: ai-4/26, ai-4/28, ai-4/31. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
| --- | --- |
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

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## Prepare, teach and assess

> Use this preparation summary alongside the detailed Metrics Match Consequences lesson; it is a checklist rather than a replacement for the full sequence.

### Teaching target

Teaching target: Apply a declared **score threshold** including equality

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Give TP, FP, FN, TN at both **threshold**s.

### Supported practice answer

Reasoned practice answer: Teacher explanation: At .5: 2, 1, 1, 2. At .3: 3, 2, 0, 1.

Check every learner before discussion: Use the worked case for **Threshold**s and **consequence**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Repair the misconception

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: At .5: A, B, C. At .3: A, B, C, D, E, including equality.

### Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

### Extend the reasoning

Extension: Miss and false-alarm costs swap to1 and 5. Require a **prediction** and a reason before comparing. Expected result: **Threshold**0.7 costs1 while0.4 costs5; preferred **threshold** switches.

Reason: Metric choice encodes the declared **consequence**s, not a universal best **threshold**.

### Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

### Project checkpoint

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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## Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery **score**s. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

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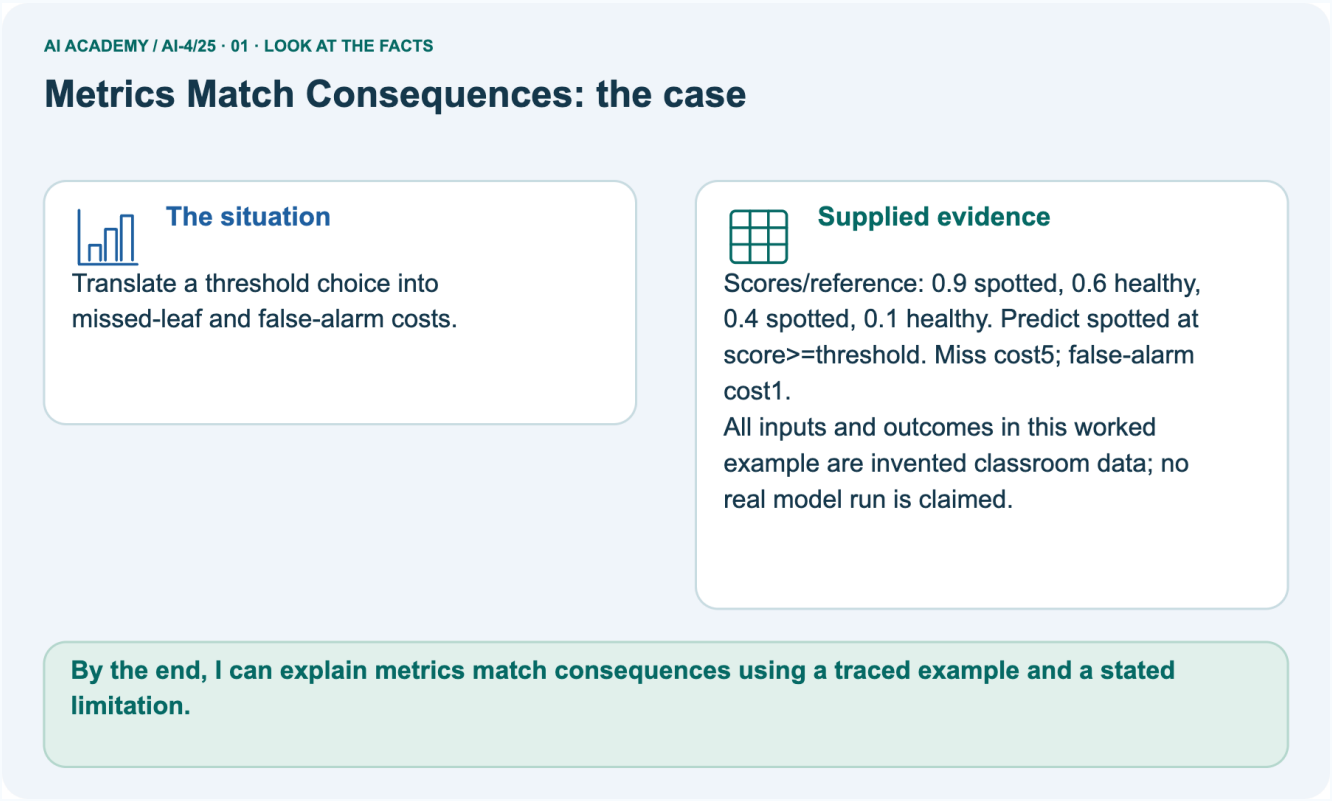


## Visual concept journey

> Show one Metrics Match Consequences graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

## 01. Metrics Match Consequences: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

Translate a **threshold** choice into missed-leaf and false-alarm costs. Supplied facts: **Score**s/reference: 0.9 spotted, 0.6 healthy, 0.4 spotted, 0.1 healthy. Predict spotted at **score**>=**threshold**. Miss cost5; false-alarm cost1.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

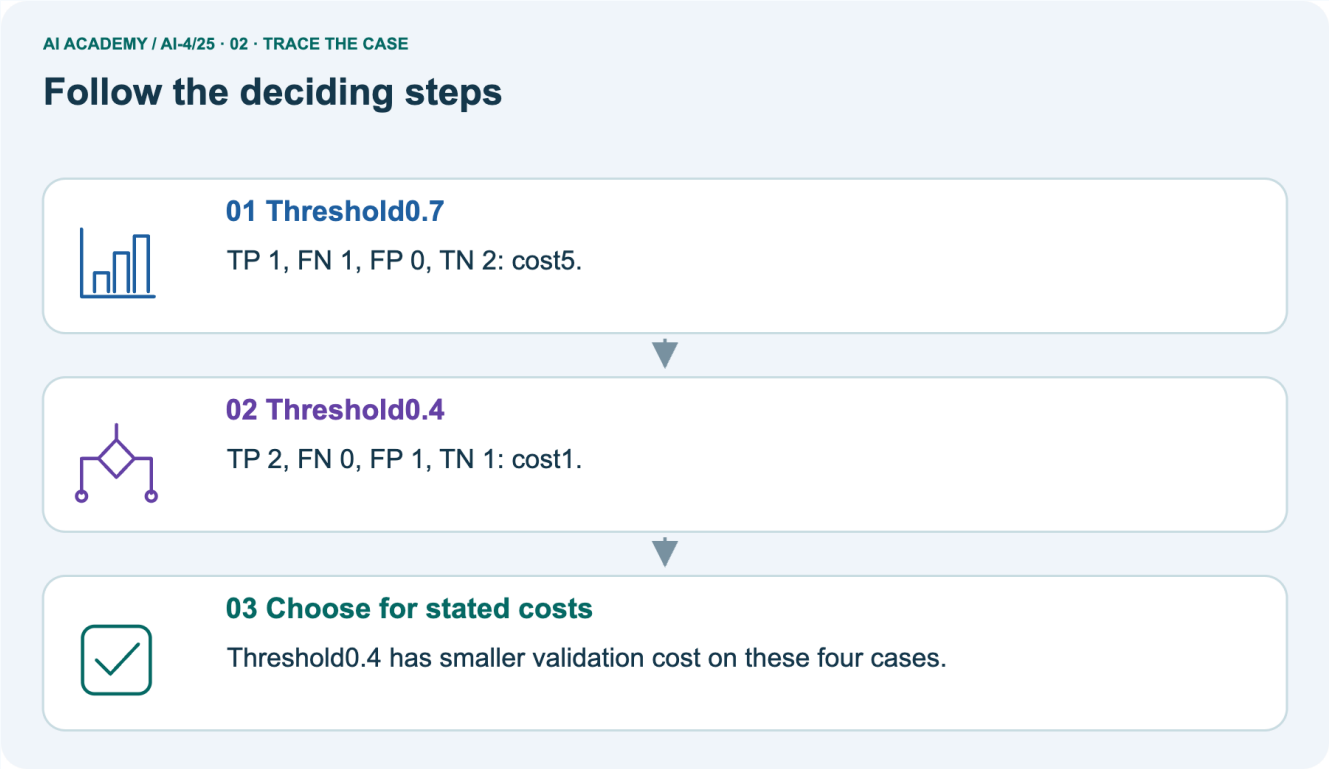
> Point to the given fact that matters most. Explain why it matters.

### Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the

image is difficult to see.

### 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

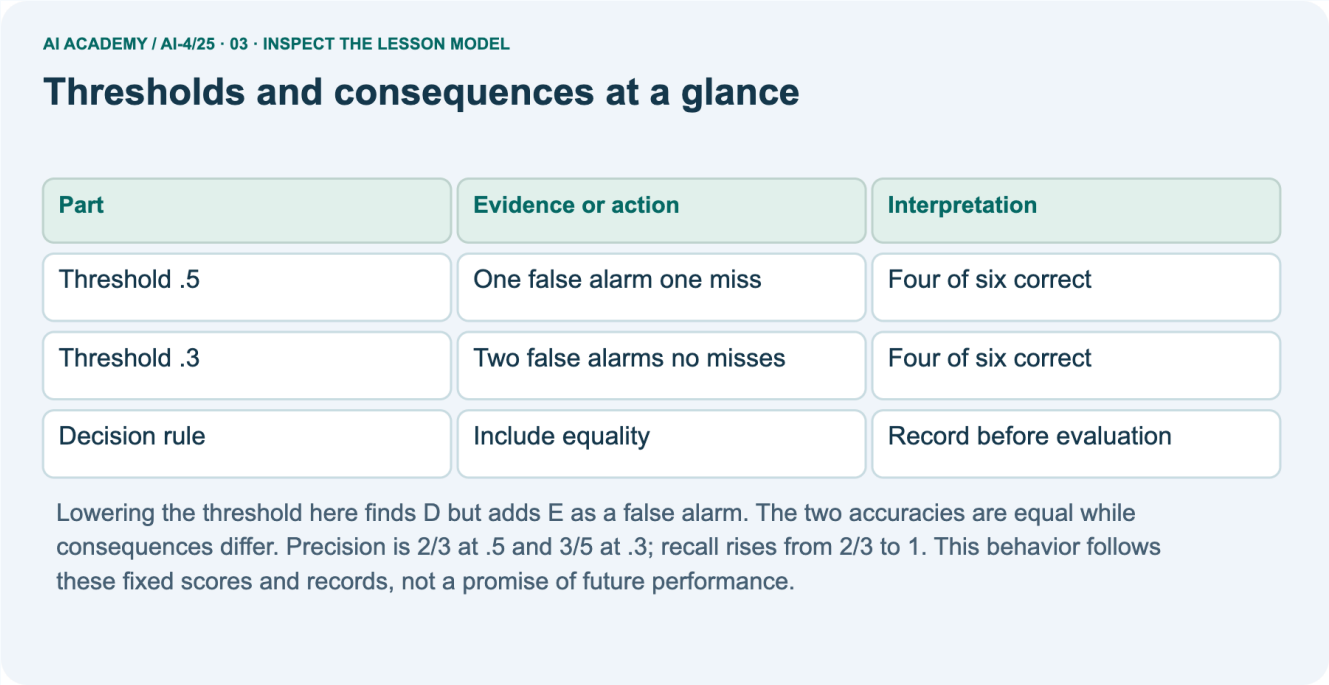
**Threshold**0.7: TP 1, FN 1, FP 0, TN 2: cost5. → **Threshold**0.4: TP 2, FN 0, FP 1, TN 1: cost1. → Choose for stated costs: **Threshold**0.4 has smaller validation cost on these four cases.

Point to each numbered stage and explain which recorded **input** it uses.

#### Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

### 03. Thresholds and consequences at a glance




Follow the labelled picture with the matching explanation. Open the vector diagram

**Threshold**s and **consequence**s at a glance. **Threshold** .5 | One **false alarm** one miss | Four of six correct. **Threshold** .3 | Two **false alarm**s no misses | Four of six correct.

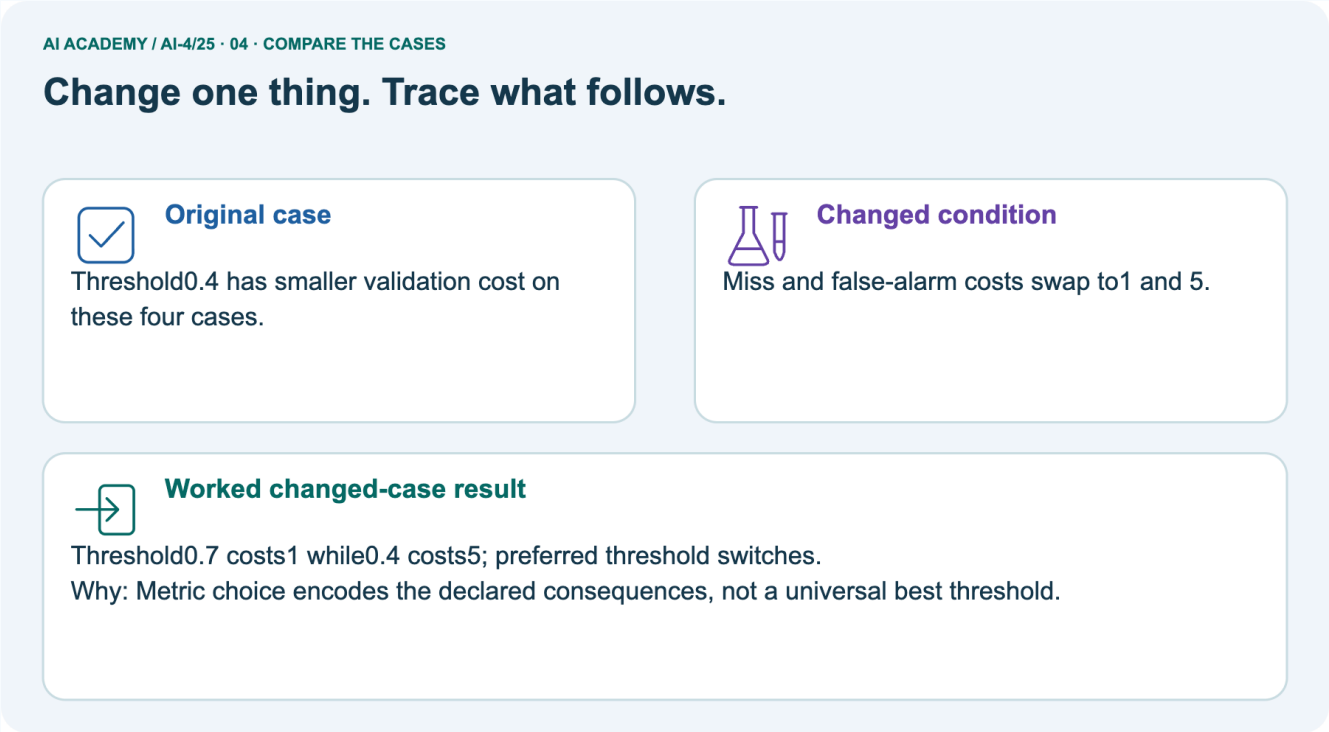
Decision rule | Include equality | Record before evaluation

> Read the labels and values. Explain the relationship the model shows.

#### Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Threshold**0.4 has smaller validation cost on these four cases. Changed condition: Miss and false-alarm costs swap to1 and 5. Expected result: **Threshold**0.7 costs1 while0.4 costs5; preferred **threshold** switches.

Reason: Metric choice encodes the declared **consequence**s, not a universal best **threshold**.

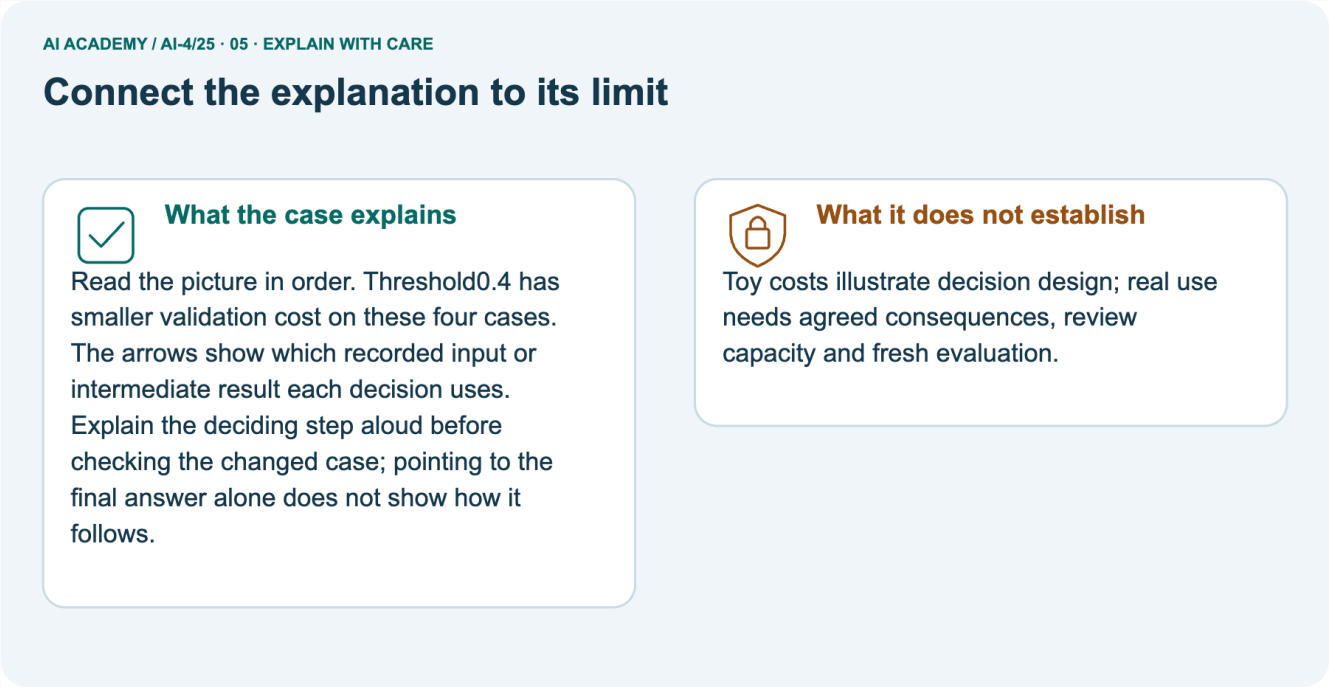
> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Metric choice encodes the declared **consequence**s, not a universal best **threshold**.

### 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. **Threshold**0.4 has smaller validation cost on these four cases. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Toy costs illustrate decision design; real use needs agreed **consequence**s, review capacity and fresh evaluation.

Explain the case, then point to a limit on the claim.

### Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Toy costs illustrate decision design; real use needs agreed **consequence**s, review capacity and fresh evaluation.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which labels determine the false-alarm count at **threshold** 0.4? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

### Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The healthy 0.6 case is wrongly flagged; healthy 0.1 stays clear. There is exactly one **false alarm**. Changed-case check: **Threshold**0.7 costs1 while0.4 costs5; preferred **threshold** switches.

Metric choice encodes the declared **consequence**s, not a universal best **threshold**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.